

Exercise 7: Financial Forecasting

Recursion is a technique where a function calls itself. It helps in implementing complex code with simple implementation. Complexity is 2^n depending upon various factors.

Code:

```
public class FinancialForecast {
    public static double forecast(double presentValue, double growthRate,
int years) {
        if (years == 0) {
            return presentValue;
        }
        return forecast(presentValue, growthRate, years - 1) * (1 +
growthRate);
    }

    public static void main(String[] args) {
        double presentValue = 100000;
        double growthRate = 0.10;
        int years = 5;
        double futureValue = forecast(presentValue, growthRate, years);
        System.out.println("Predicted Future Value = " + futureValue);
    }
}
```

Output:

```
1 public class FinancialForecast {
2     public static double forecast(double presentValue, double growthRate, int years) {
3         if (years == 0) {
4             return presentValue;
5         }
6         return forecast(presentValue, growthRate, years - 1) * (1 + growthRate);
7     }
8
9     Run | Debug
10    public static void main(String[] args) {
11        double presentValue = 100000;
12        double growthRate = 0.10;
13        int years = 5;
14        double futureValue = forecast(presentValue, growthRate, years);
15        System.out.println("Predicted Future Value = " + futureValue);
16    }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\adits\Desktop\DeepSkillig\Coding> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\adits\AppData\Roaming\Code\User\workspaceStorage\1a3385de95c309e825d680cb68353935b\redhat.java\jdt_ws\Coding_613b24a0\b' in 'FinancialForecast'
Predicted Future Value = 161051.0000000001
PS C:\Users\adits\Desktop\DeepSkillig\Coding>
```

Complexity analysis:

There is only 1 recursive call per year hence the height of the recursive tree is n and width is 1. Time complexity is $n \times 1$ i.e. $O(n)$